Wen-Hsi Kuo

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Conservation Genomics Postdoctoral Fellow
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EDUCATION & Appointments

2024 – present	Conservation Genomics Postdoctoral Fellow	**Missouri Botanical Garden** Center for Conservation and Sustainable Development
2018 – 2024	Ph.D. Ecology & Evolution	**Washington University in St. Louis** Department of Biomedical & Biological Science
2016-2018	Research Assistant	**Biodiversity Research Center** Academia Sinica, Taiwan
2012-2015	M.S. Ecology & Evolution	**National Taiwan University** Institute of Ecology & Evolution
2008-2012	B.S. Life Science	**National Taiwan University** Department of Life Science

PUBLICATIONS

Kuo, W. H., Landon, C., Bassüner, B., Linan, A. G., Littlefield, T. R., Albrecht, M. A., Edwards, C. E. (2026). Range-wide population genetics reveals how disjunctions, admixture, and translocations influence conservation of *Physaria globosa* (Brassicaceae). Under review.

Kuo, W. H., Aguirre, E, Bassüner, B., Coble, E., Albrecht, M. A., Delfeld, B., Dell, N., Engelhardt, M., Hathaway, J., Jones, M. M., Sucher, R., Edwards, C. E. (2026). Genomic consequences of altered flood regimes: gene flow and adaptation in a threatened floodplain plant. Under review.

Kuo, W. H., Small, L, Olsen, K. M. (2026). Phenotypic and genetic bases of variable drought stress responses in a widely adapted allotetraploid species. American Journal of Botany. 113: e70200. doi: 10.1002/ajb2.70200

Kuo, W. H., Cunningham, E. (mentored undergrad), Guo, E. (mentored undergrad), Olsen, K. M. (2024). Genetics and plasticity of white leaf mark variegation in white clover (*Trifolium repens* L.). Annals of Botany, 134(6): 949-958. doi: 10.1093/aob/mcae129

Kuo, W. H., Zhong, L., Wright, S. J., Goad, D., Olsen, K. M. (2024). Beyond cyanogenesis: Temperature gradients drive environmental adaptation in North American white clover (*Trifolium repens* L.). Molecular Ecology, 33(17): e17484. doi: 10.1111/mec.17484

Kuo, W. H., Wright, S. J., Small, L, Olsen, K. M. (2024). *De novo* genome assembly of white clover (*Trifolium repens* L.) reveals that copy number variation contributes to rapid climatic adaptation. BMC Biology, 22(1):165. doi: 10.1186/s12915-024-01962-6

Kuo, W. H., Small, L, Olsen, K. M. (2023). Variable expression of cyanide detoxification and tolerance genes in cyanogenic and acyanogenic white clover (*Trifolium repens* L.). American Journal of Botany, 110(10): e16233. doi: 10.1002/ajb2.16233

Santangelo, J. S., Battlay, P., Hendrickson, B. T., **Kuo, W. H.**, Olsen, K. M., Kooyers, N. J., Johnson, M. T. J., Hodgins, K. A., Ness, R. W. (2023). Haplotype-Resolved, Chromosome-Level Assembly of White Clover (*Trifolium repens* L., Fabaceae). Genome Biology and Evolution, 15(8): 2023.06.06.543960. doi: 10.1093/gbe/evad146

Kuo, W. H., Liu, S. H., Chang, C. C., Hsieh, C. L., Li, Y. H., Ito, T, Won, H, Kokubugata, G., Chung, K. F. (2022). Plastome phylogenomics of *Allaeanthus*, *Broussonetia* and *Malaisia* (Dorstenieae, Moraceae) and the origin of *B.* × *kazinoki*. Journal of plant research, 135(2): 203-220. doi: 10.1007/s10265-022-01369-w

Pan, Z. J., Nien, Y. C., Shih, Y. A., Chen, T. Y., Lin, W. D., **Kuo, W. H.**, Hsu, H. C., Tu, S. L., Chen, J. C., Wang, C.N. (2022). Transcriptomic Analysis Suggests Auxin Regulation in Dorsal-Ventral Petal Asymmetry of Wild Progenitor *Sinningia speciosa*. International Journal of Molecular Sciences, 23(4): 2073. doi: 10.3390/ijms23042073

Kokubugata, G., Nakamura, K., **Kuo, W.H**., Qi, Z.C., Chung, K.F., Fu, C.X., Suzuki, Y., Yokota, M. (2019). Reappraisal of *Tashiroea* as a genus independent of *Bredia* (Melastomataceae) based on molecular data. Phytotaxa, 392(1): 75-83. doi: 10.11646/phytotaxa.392.1.8

Kuo, W. H., Hung, Y. L. (mentored graduate), Wu, H. W. (mentored undergrad), Hong, C. Y., Wang, C. N. (2018). Shoot regeneration process and optimization of *Agrobacterium*-mediated transformation in *Sinningia speciosa*. Plant Cell, Tissue and Organ Culture, 134(2): 301-316. doi:10.1007/s11240-018-1424-7

Hsu, H.J., He, C.W. (mentored graduate), **Kuo W.H.**, Hsin K.T., Lu, J.Y., Pan, Z.J., Wang, C.N. (2018). Genetic Analysis of Floral Symmetry Transition in African Violet Suggests the Involvement of *Trans*-acting Factor

for *CYCLOIDEA* Expression Shifts. Frontiers in Plant Science, 9: 1008. doi: 10.3389/fpls.2018.01008

Chung, K. F., **Kuo, W. H.**, Hsu, Y. H., Li, Y. H., Rubite, R. R., & Xu, W. B. (2017). Molecular recircumscription of *Broussonetia* (Moraceae) and the identity and taxonomic status of *B. kaempferi* var. *australis*. Botanical Studies, 58: 11. doi:10.1186/s40529-017-0165-y

Penailillo, J., **Kuo, W. H.**, Olivares, G., Silva-Poblete,G., Pena-Ahumada, B., Munoz, S., Moncada X., Chung, K. F., Seelenfreund, D., Seelenfreund, A. (2017). Characterization of microsatellites for *Broussonetia papyrifera* (Moraceae). Applications in Plant Sciences, 5(8): 1700044. doi: 10.3732/apps.1700044

GRANTS, FELLOWSHIPS AND AWARDS

Postdoctoral Fellowship, Missouri Botanical Garden	2024-2026
Botanical Society of America Hardship Travel Award (USD 1,400)	2025
William H. Danforth Plant Sciences Fellowship (stipend and research funding, USD 130K)	2019-2023
Ministry of Education Studying Abroad Scholarship (Research funding, USD 32K)	2021-2022
NIGINTERN Program fellowship, National Institute of Genetics, Japan (USD 2,260)	2011

TEACHING AND MENTORSHIP

Graduate Teaching

Evolutionary Biology - *Discussion instructor* -*Teaching Assistant*	Washington University in St. Louis	2019, 2023
General Biology - *Teaching Assistant*	National Taiwan University	2013-2014
Botany lab - *Instructor* - *Teaching Assistant*	National Taiwan University	2012-2013

Research Mentorship (1 – undergraduate student; 2 – graduate student; 3 – intern)

Maia Jones[2]	Missouri Botanical Garden	2024 – present
John Yocum[3]	Missouri Botanical Garden	2026 summer
Ethan Coble[3]	Missouri Botanical Garden	2025 summer
Megan Connor[3]	Missouri Botanical Garden	2025 summer
Maia Washington[3]	Missouri Botanical Garden	2024 summer
Sage Waite[3]	Missouri Botanical Garden	2024 summer

Raymond Castillo[2]	Washington University in St. Louis	2023
Emily Guo[1]	Washington University in St. Louis	2023. Published. doi: 10.1093/aob/mcae129
Eimear Cunningham[1]	Washington University in St. Louis	2021. Published. doi: 10.1093/aob/mcae129
Nicholas Ho[1]	Washington University in St. Louis	2021
Grace Li[1]	Washington University in St. Louis	2020
Cheyenne Anderson[2]	Washington University in St. Louis	2020
Chia-Rong Chuang (莊嘉瑢)[3]	Academia Sinica, Taiwan	2017
Hsin-Hua Stacy Chuang (莊欣樺)[3]	Academia Sinica, Taiwan	2016
Cheng-Wen He (何承紋)[2]	National Taiwan University	2013-2015. Published. doi: 10.3389/fpls.2018.01008
Yu-Ling Hung (洪育翎)[2]	National Taiwan University	2014-2015. Published. doi:10.1007/s11240-018-1424-7
Ho-Wei Wu (吳和為)[1]	National Taiwan University	2013-2015. Published. doi:10.1007/s11240-018-1424-7
王心柔 [1]	National Taiwan University	2013-2014

ACADEMIC PRESENTATIONS

Invited Seminars

Hokkaido University, Japan	University-wide open seminar	2026
National Taiwan University, Taiwan	Candidate seminar	2026
National Taiwan University, Taiwan	Lab-invited seminar	2024
Washington University in St. Louis	LEC/EEB seminar	2022, 2023 (twice), 2024
Washington University in St. Louis	Bioforum series	2020

Conference contributions (1 – Oral presentation; 2 – Poster)

Botanical Society of America Botany Conference[1]	2026
St. Louis Ecology / Evolution / Conservation (SLEEC)[1]	2025
Botanical Society of America Botany Conference[1]	2025
Center For Plant Conservation National Meeting[1]	2025
MBG Informatics Symposium[1]	2024
Botanical Society of America Botany Conference[1]	2022
St. Louis Ecology / Evolution / Conservation (SLEEC)[2]	2019
Taiwan-Japan Plant Biology Conference[2]	2017

Taiwan Society of Plant Biologists[2] 2015

NTU & NTNU Joint Symposium on Ecology and Evolutionary Biology[12] 2014

SERVICE

Manuscript referee Total 12 papers

Journal of Heredity (3), Environmental and Experimental Botany (1),
Molecular Ecology (2), Peer J (1), Plos One (2), BMC Plant Biology (1).
Evolutionary Ecology (1), Conservation Genetics (1)
https://www.webofscience.com/wos/author/record/HOH-0053-2023

Washington University EEPB Seminar Series 2022

Role: Organizer & Host

PROFESSIONAL REFERENCES

Dr. Kenneth Olsen
Ph.D. advisor

George William and Irene Koechig Freiberg Professor of Biology
Associate Chair of Climate and Facilities
Washington University in St. Louis, MO, USA
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Dr. Christine Edwards
Postdoctoral advisor

Honorary Adjunct Professor
Stephen and Camilla Brauer Associate Scientist
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Dr. Kuo-Fang Chung
Research assistant advisor

Museum Director & Research Fellow
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Dr. Chun-Neng Wang
M.S. advisor

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